



ISLAMIC UNIVERSITY OF TECHNOLOGY (IUT)

ORGANISATION OF ISLAMIC COOPERATION (OIC)

Department of Computer Science and Engineering (CSE)

Project Report

SWE 4304: Software Project Lab I

Project Title

StudyPet – Your Virtual Study Companion

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Project Overview

This project is a Command-Line Interface (CLI) based Study Companion Pet system designed to help students maintain consistent study habits through gamification and emotional engagement. The system allows users to create an account, set daily study goals, and interact with a virtual pet that reflects the user's study behavior, mood, and consistency.

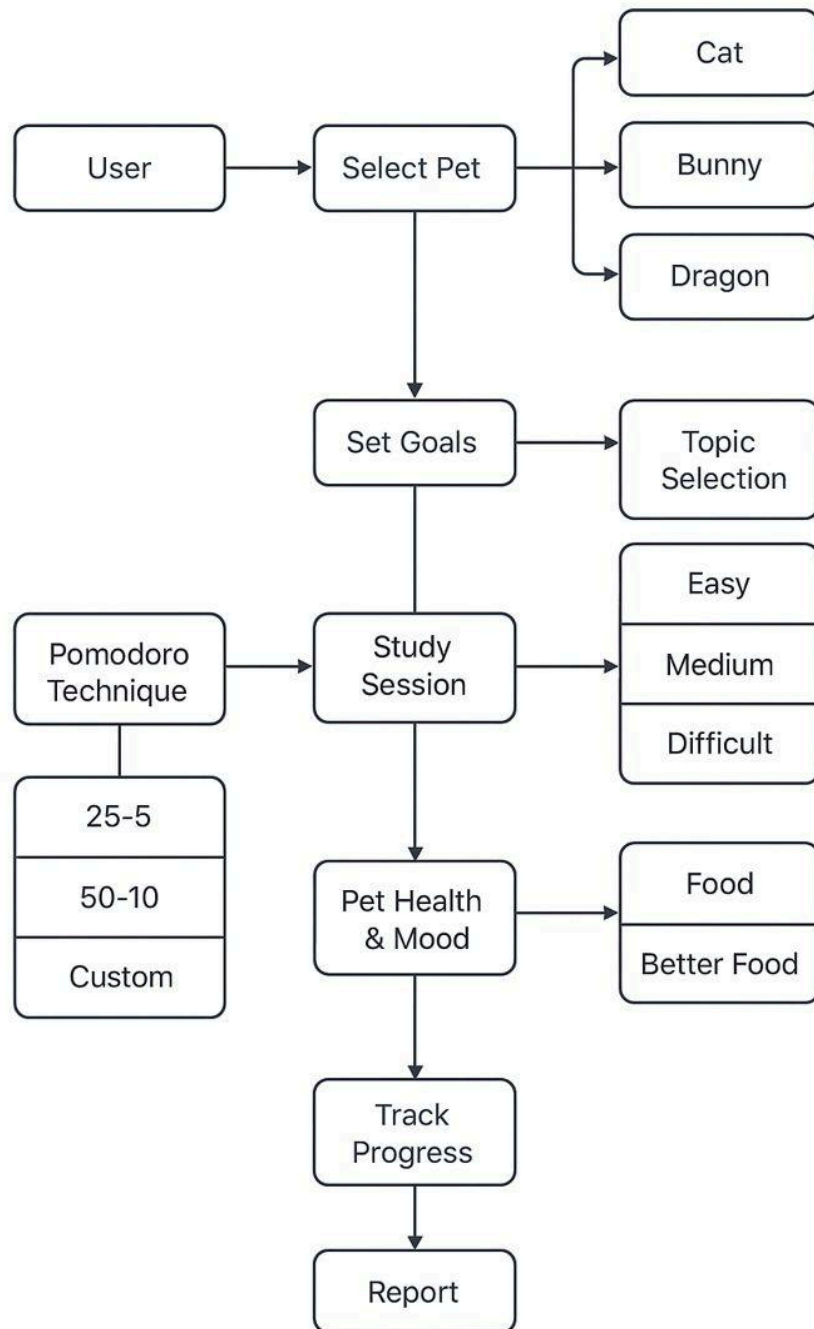
The application integrates study tracking techniques, including Pomodoro sessions, goal-based performance analysis, and health-based virtual pet management. By combining productivity tools with a reward-based system, the project encourages discipline, self-awareness, and motivation among students.

Motivation Behind the Project

Many students struggle with maintaining consistency, motivation, and emotional balance while studying. Traditional productivity tools often lack emotional feedback and engagement. This project is motivated by the idea of transforming studying into an interactive experience where progress, mood, and effort directly affect a virtual companion.

By introducing a pet that reacts to the user's habits and emotions, the system aims to make studying more engaging, accountable, and rewarding, while also promoting healthy study routines and self-reflection.

System Flow Chart



Key Features

1. Account Creation and Personalization

Users create an account by providing their name, daily study hours, and desired academic goals. Based on this information, the system assigns a personality type to the pet: Lazy, Mediocre, or Serious. Users can also choose a pet theme, such as Cat, Bunny, or Dragon.

2. Coin and Health System

The system initializes with Health = +10 and Coins = 5. Completing study sessions earns coins, which can be used to feed the pet. Pet health reflects the user's consistency and mood. If health drops below a threshold, warning messages are shown.

3. Topic Selection and Difficulty Levels

Users can select study topics and choose difficulty levels:

- Easy: 5 coins
- Medium: 10 coins
- Hard: 20 coins

The chosen difficulty affects coin rewards and pet emotions during the study session.

4. Pomodoro-Based Study Sessions

The system supports predefined Pomodoro sessions:

- 25 minutes study / 5 minutes break
- 50 minutes study / 10 minutes break
- Customizable sessions

During sessions, the pet shows random animations influenced by the user's mood and topic difficulty.

5. Pet Shop System

Users can purchase food items to restore pet health:

- Normal Food: 50 coins
- Premium Food: 75 coins
- Better food results in greater health recovery.

6. Mood Tracking and Emotional Feedback

On launching the application, users input their daily mood and study goals. Based on mood input, the pet displays encouraging messages.

7. Inactivity Penalty System

If the user does not log in for seven consecutive days, the pet is considered dead, and the user will have to start over with a new one. Upon the next login, the system displays Health = 10 and Coins = -100, serving as a strong penalty and reminder.

Advanced Features:

• Study Analytics and Visualization

The system provides analytical insights into the user's study behavior over time. A GitHub-style heatmap is used to visually represent daily study intensity, allowing users to quickly identify patterns of consistency, inactivity, or improvement across days. This visual representation helps users self-evaluate their discipline and long-term engagement.

• Quiz Marks and Performance Prediction

Users can input quiz marks to evaluate how much of the syllabus has been covered and how much remains. The system estimates future performance based on current study patterns.

● **Weekly Performance Report**

The system automatically generates a 7-day performance report, which summarizes:

- Number of Pomodoro sessions completed
- Goal achievement rate
- Mood trends
- Pet health status
- Overall productivity level

This report allows users to reflect on their weekly performance and plan improvements for the following week.

● **Mood-Based Health Impact**

User mood is tracked daily to reflect emotional well-being. If the mood is recorded as “Tired” for five consecutive days, the pet’s health decreases, and the pet displays signs of sickness. This feature emphasizes the importance of mental health and balanced study habits.

● **Recreation and Stress-Relief Unlock System**

To promote healthy breaks and prevent burnout, the system unlocks recreational options after completing five or more Pomodoro sessions in a day. These options include:

- Guided meditation
- Simple CLI-based mini-games

This mechanism rewards productivity while encouraging relaxation and stress reduction.

Tools and Technologies

- **Programming Language:** Python
- **Interface:** Command Line Interface (CLI)
- **Development Environment:** Visual Studio Code (VS Code)
- **Data Handling:** File Handling in Python
- **Version Control:** Git (for code backup and progress tracking)

Proposed Timeline

Week 1 — Requirement analysis & project planning

Week 2 — System design, architecture & flow diagram

Week 3 — Core framework & basic feature setup

Week 4 — Account creation & user profile system

Week 5 — Pet core mechanics (health, energy, mood)

Week 6 — Topic-based study & timer system

Week 7 — Feeding system & reward logic

Week 8 — Pet shop features & animations

Week 9 — Mood tracking & health logic

Week 10 — Login streaks & goal tracking along with testing

Week 11 — Heatmap & study activity tracking along with testing

Week 12 — Reports & quiz/feedback input along with testing

Week 13 — Prediction models & insights along with testing

Week 14 — Final testing, deployment & submission

Action Taken Based on the Feedback

Initially, the project was planned to use a database management system (PostgreSQL) for storing user data, study records, and pet status. However, based on the faculty's feedback, the use of a database was marked as optional and not required for the scope of this SPL project.

Following this guidance, the system design was revised to use file handling in Python for data storage instead of a database. This change simplified the implementation, reduced system complexity, and ensured better alignment with the course requirements while still maintaining data persistence and functionality.

Conclusion

The StudyPet system provides an engaging and structured approach to studying by blending productivity techniques with emotional and reward-based feedback. The project emphasizes consistency, self-awareness, and motivation, making it a practical and interactive tool for students.